

Capstone Project-1 Report

“AI Workforce Impact Analysis Dashboard”

Domain: Artificial Intelligence & Workforce Analytics

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1. Introduction

Artificial Intelligence is transforming industries worldwide by automating tasks, improving efficiency, and reshaping workforce requirements. While AI creates new opportunities, it also poses challenges related to job displacement and skill adaptation.

This project focuses on analyzing the impact of AI on different industries, countries, and job roles. An interactive Power BI dashboard was developed to help organizations understand AI adoption trends, workforce risks, and future employment opportunities.

2. Problem Statement

Organizations and governments need to understand how AI affects employment across different sectors. Some jobs face a higher risk of automation, while others experience increased demand due to technological advancements.

Without proper analysis, decision-makers may struggle to identify vulnerable job roles, plan workforce development strategies, and prepare employees for future skill requirements.

3. Objective

The primary objectives of this project are:

- Analyze workforce trends influenced by AI adoption.
 - Identify industries and job roles with high automation risk.
 - Measure AI adoption levels across countries and sectors.
 - Visualize key workforce metrics using interactive dashboards.
 - Support strategic workforce planning and policy-making.
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4. Tools & Technologies

Tool	Purpose
Power BI	Dashboard Development and Reporting
Excel/CSV	Data Source
DAX	KPI Calculations and Measures

5. Methodology

1. Data Collection

- Imported AI Imact on jobs 2030 dataset From Kaggle.

- Verified data quality and consistency.

2. Data Cleaning

- Removed duplicate records.
- Handled missing values.
- Standardised job role and industry names.

3. Data Analysis

- Calculated AI adoption percentages.
- Analysed automation risk by role.
- Compared workforce trends across countries.
- Most effected roles and employees

4. Dashboard Development

- Created KPI cards.
 - Developed charts and visualisations.
 - Added slicers and interactive filters.
 - Designed multiple dashboard pages for detailed analysis.
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6. Data Description

Dataset contains workforce and AI-related information including:

- Industry
- Country
- Job Role
- AI Adoption Level
- Automation Risk
- Employees Affected
- Salary Impact
- Workforce Demand
- Future Growth Potential

These attributes help evaluate AI's influence on employment and industry transformation.

7. Dashboard Design

Dashboard 1:AI Workforce Analytics

Displays:

- Key Performance Indicators[KPI]
- Slicers
- Top5 AI Replacement risk roles and Industries through column chart and bar charts
- Feature Job Demand score Roles
- Employees risk and Demand table

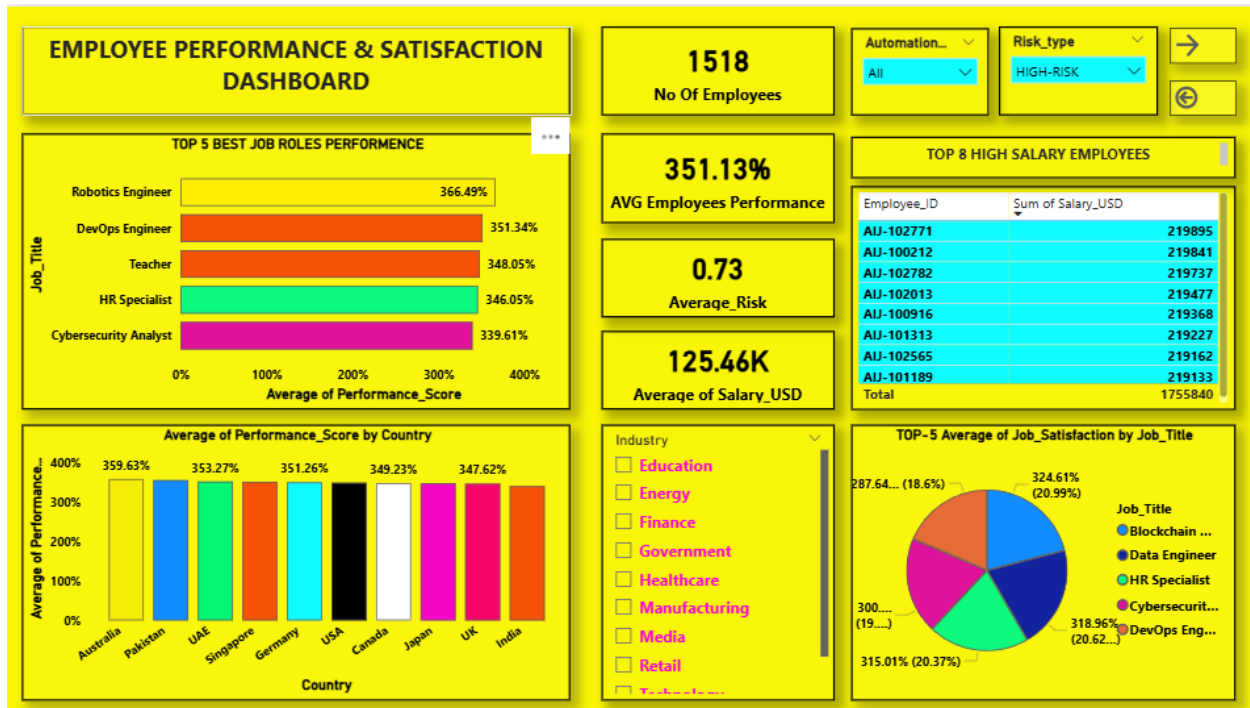


Dashboard 2: Employees Performance and Satisfaction Analysis

Displays:

- KPI's
- Country-wise Employees performance Analysis

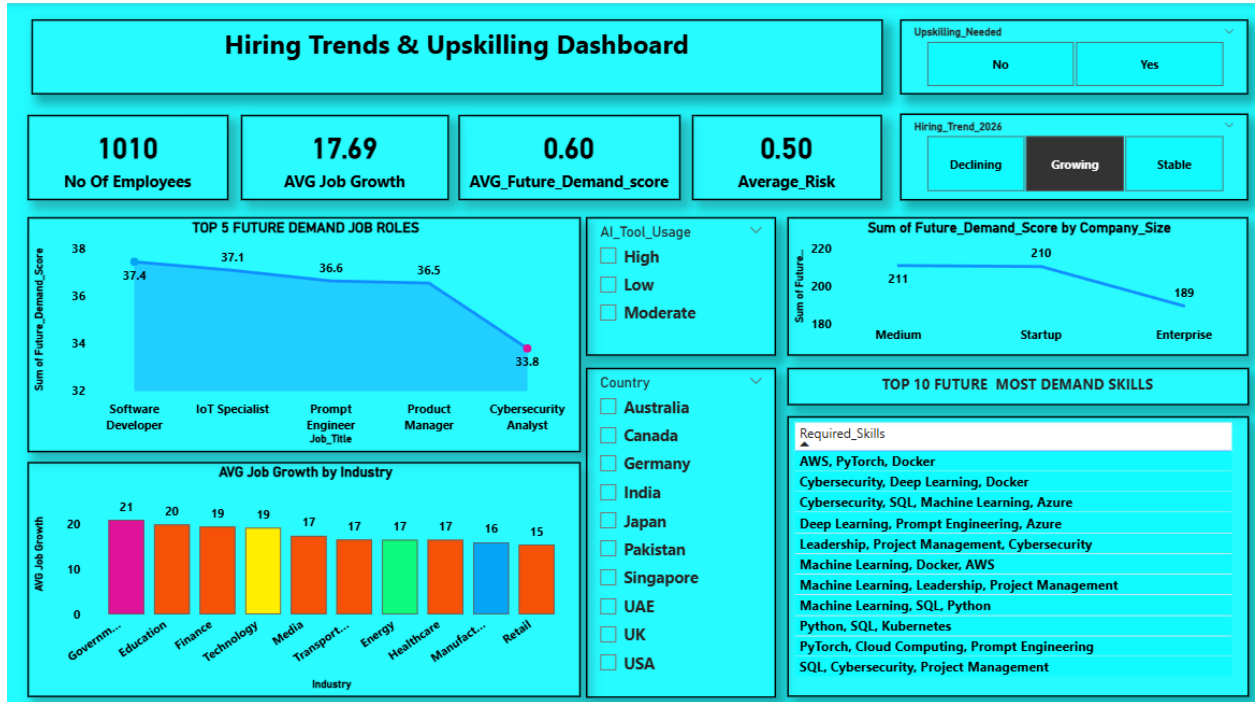
- Best Performing Roles
- Workforce Demand Trends
- Employment Distribution
- Top 5 average job satisfaction job roles
- Highest salary Earning Employees



Dashboard 3: Hiring Trend and Upskilling Analysis

Displays:

- KPI's
- Top future-demand job roles
- Job Growth By industries
- Feature Demand score distribution by Company Size
- Top 10 Feature Demand skills
- Slicers



4: Detailed Insights

Displays:

- Industry Breakdown
- Country Breakdown
- Role-wise Analysis
- Interactive Filtering

INSIGHTS

==>The top five job roles expected to face high AI replacement risk (above 60%) are: [DevOps Engineer, Cybersecurity Analyst, Teacher, HR Specialist, Data Engineer]

==>Among these roles, DevOps Engineer has the highest AI replacement risk at 70%, with 98 employees currently working in this role.

==>The Government industry and Pakistan are projected to experience the highest AI replacement risk compared to other industries and countries.

==>The job role with the highest future demand without requiring upskilling is Software Engineer.

==>The job role with the highest future demand after upskilling is Data Scientist.

==>In 2026, the role showing the strongest hiring trend is Software Developer.

==>Employees working as DevOps Engineers, Cybersecurity Analysts, Teachers, HR Specialists, and Data Engineers are encouraged to learn AI tools and AI-related skills, as these roles are among the most vulnerable to AI-driven automation.

Conclusion

The analysis suggests that while several job roles face significant AI replacement risk, professionals who adopt AI tools and continuously upskill can improve their career prospects and remain competitive in the evolving workforce.

8. Key Insights

1. DevOps Engineer shows one of the highest AI replacement risks.
2. Cyber Security Analyst and Data Engineer roles face significant automation exposure.
3. Government sector demonstrates considerable workforce impact.
4. Certain countries exhibit higher AI adoption rates compared to others.
5. AI adoption is positively correlated with productivity improvements.
6. Workforce demand is shifting toward AI-related and analytical roles.
7. Several traditional roles require upskilling to remain competitive.

9. Challenges Faced

- Handling missing and inconsistent data values.
- Designing meaningful KPIs.
- Creating interactive visualisations.

- Establishing relationships between multiple attributes.
 - Optimising dashboard performance.
 - Ensuring accurate DAX calculations.
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10. Conclusion

The AI Workforce Impact Analysis Dashboard provides valuable insights into how artificial intelligence is reshaping industries, countries, and job roles. Through interactive visualisations and data-driven analysis, organisations can better prepare for workforce transformation and make informed strategic decisions.

The dashboard serves as an effective tool for understanding automation risks, workforce trends, and future employment opportunities in an AI-driven economy.

11. Future Scope

1. Predictive Workforce Modeling.
 2. Real-Time AI Adoption Monitoring.
 3. Skill Gap Analysis.
 4. Employee Reskilling Recommendations.
 5. Industry-Specific Forecasting.
 6. Integration with Live Employment Data.
 7. AI-Based Career Guidance System.
 8. Advanced Workforce Demand Forecasting.
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12. References

1. Kaggle Workforce Analytics Datasets
2. Microsoft Power BI Documentation
3. Artificial Intelligence Workforce Research Reports
4. Industry Employment Trend Reports